

Practical: Create a Docker File for a Web Application and Host It on AWS

Objective: To create and implement a Dockerfile for a self-created web application, build a Docker image, run the application inside a Docker container, and host it on AWS.

Requirements

- AWS Account
- Docker Desktop
- AWS EC2 instance
- Basic HTML/Python/Flask knowledge
- SSH client

Suggested Application

Create a simple Flask web application.

```
docker-web-app/  
├──  
│   ├── app.py  
│   ├── requirements.txt  
│   ├── Dockerfile  
│   └── templates/  
│       └── index.html
```

1. Create app.py

```
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    return "<h1>Welcome to My Docker Web Application</h1><p>Hosted on AWS EC2</p>"  
  
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

2. Create requirements.txt

```
Flask
```

3. Create the Dockerfile

```
FROM python:3.11-slim  
  
WORKDIR /app  
  
COPY requirements.txt .  
  
RUN pip install --no-cache-dir -r requirements.txt  
  
COPY . .  
  
EXPOSE 5000  
  
CMD ["python", "app.py"]
```

4. Build Docker Image

```
docker build -t my-web-app .
docker images
```

5. Run the Container

```
docker run -d -p 5000:5000 --name web-container my-web-app
docker ps
```

```
# Test locally:
http://localhost:5000
```

6. Create an AWS EC2 Instance

1. Open EC2 in AWS Management Console.
2. Click Launch Instance.
3. Select Ubuntu Server.
4. Select an appropriate instance type.
5. Create/select a key pair.
6. Configure the security group.
7. Allow SSH (TCP 22).
8. Allow Custom TCP (TCP 5000).
9. Launch the instance.

7. Connect to EC2

```
ssh -i "your-key.pem" ubuntu@<EC2-PUBLIC-IP>
```

8. Install Docker on EC2

```
sudo apt update
sudo apt install docker.io -y
docker --version
sudo systemctl start docker
sudo systemctl enable docker
```

9. Transfer the Application to EC2

```
git clone <your-github-repository>
cd docker-web-app
```

10. Build the Docker Image on AWS

```
sudo docker build -t my-web-app .
sudo docker images
```

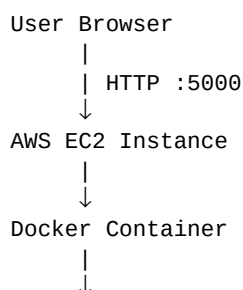
11. Run the Docker Container on AWS

```
sudo docker run -d -p 5000:5000 --name web-container my-web-app
sudo docker ps
```

12. Access the Web Application

```
http://<EC2-PUBLIC-IP>:5000
```

Expected Architecture



Result

The self-created Flask web application was successfully containerized using Docker and hosted on an AWS EC2 instance. The application was accessed through the EC2 public IP address and exposed Docker port 5000.

Viva Questions

1. What is Docker?
2. What is a Dockerfile?
3. What is the difference between a Docker image and a container?
4. Why is EXPOSE 5000 used?
5. Why do we use 0.0.0.0 in Flask?
6. What is the purpose of docker build?
7. What is the purpose of docker run?
8. Why must port 5000 be allowed in the EC2 security group?
9. What is an EC2 instance?
10. What are the advantages of containerizing an application?